

2.7.28

EE25BTECH11016 - Taraka Abhinav

Question:

Find the area of the triangle whose vertices are $A(2, 3)$, $B(3, 5)$, $C(4, 4)$.

Solution:

Point	Vector
A	$\begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$
B	$\begin{bmatrix} 3 \\ 5 \\ 0 \end{bmatrix}$
C	$\begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix}$

TABLE I: Vectors

$$(\mathbf{A} - \mathbf{B}) = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} - \begin{bmatrix} 3 \\ 5 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 0 \end{bmatrix}, \quad (1)$$

$$(\mathbf{A} - \mathbf{C}) = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} - \begin{bmatrix} 4 \\ 4 \\ 0 \end{bmatrix} = \begin{bmatrix} -2 \\ -1 \\ 0 \end{bmatrix}. \quad (2)$$

Using (1) and (2) The magnitude of the cross product is

$$\|(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{C})\| = \sqrt{0^2 + 0^2 + (-3)^2} = 3. \quad (3)$$

Therefore the area of triangle ABC is

$$\text{ar}(\triangle ABC) = \frac{1}{2} \|(\mathbf{A} - \mathbf{B}) \times (\mathbf{A} - \mathbf{C})\| = \frac{1}{2} \times 3 = 1.5. \quad (4)$$

Therefore $\text{ar}(\triangle ABC) = 1.5$.

